University of Kentucky Emergency Department	Guideline #
Title/Description: Intranasal Dexmedetomidine Guideline	
Purpose: To provide guidelines for administration of intranasal Dexmedetomidine for anxiolysis and/or agitation in the Pediatric Emergency Department	

Procedure

Indications
- Anxiolysis in patients with behavioral agitation and acute psychosis

Contraindications and Exclusion Criteria
- Allergy or sensitivity to the medication
- Difficult airway
- Nasal obstruction or trauma
- Epistaxis
- Rhinitis
- Medical conditions that affect ciliary function (ex: cystic fibrosis)
- Patients < 6 months of age
- Relative contraindications: asthma
- Patients > 100 kg

Concomitant or recent doses of opioids or benzodiazepines require increased monitoring due to potentiation of sedative effects. If agents are given closely together (less than 30 minutes) documentation for a procedural sedation should be considered.

Ordering
- Physicians may order intranasal Dexmedetomidine if none of the contraindications or exclusion criteria noted above are present.
- Contact ED pharmacy for medication.

Dosing

	Dexmedetomidine
Onset	25 – 45 minutes
Duration	45 – 120 minutes
Concentration	100 mcg/mL
Dosing	2 – 5 mcg/kg Max dose 200 mcg Divided between each nostril, max volume per nostril is 1 mL
Monitoring	Heart rate, blood pressure, pulse-oximetry Assess vital signs every 15 minutes for 1 hour post administration. Vitals should be assessed again prior to discharge

Supplies

The mucosal atomizer device (MAD) nasal adaptor will be used with a syringe. This atomizer delivers a metered dose of drug into the nares. Draw up doses with 0.1mL of excess to account for dead space in the atomizer.

Documentation

Follow standard ED documentation for anxiolysis

Adverse Events

Cardiovascular events (bradycardia and hypotension)
Respiratory Depression

References:

1. Yuen VM, Hui TW, Irwin MG, Yuen MK. A comparison of intranasal dexmedetomidine and oral midazolam for premedication in pediatric anesthesia: a double-blinded randomized controlled trial. Anesth Analg. 2008;106(6):1715-1721.
2. Yuen VM, Hui TW, Irwin MG, et al. A randomized comparison of two intranasal dexmedetomidine doses for premedication in children. Anaesthesia. 2012;67(11):1210-1216. doi:10.1111/j.1365-2044.2012.07309.x
3. Neville DN, Hayes KR, Ivan Y, McDowell ER, Pitetti RD. Double-blind Randomized Controlled Trial of Intranasal Dexmedetomidine Versus Intranasal Midazolam as Anxiolysis Prior to Pediatric Laceration Repair in the Emergency Department. Acad Emerg Med. 2016;23(8):910-917.

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